

A Review of Microgrids and A Study on Hybrid Renewable Energy Microgrid(PV-Battery Microgrid and PV-Hydrogen-Battery Microgrid)

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Abstract: This study aims to design and model a renewable energy system (RES) microgrid. First, we will model a simple grid-connected microgrid with a PV array, battery, load, and a grid represented as an AC voltage source in Simulink Matlab. The PV array is connected to a boost converter and equipped with Maximum Power Point Tracking (MPPT) Control. We initially model various solar irradiance using different function blocks (such as step, sine, or ramp) blocks in Simulink. The battery is connected to the microgrid with a bidirectional converter. Afterwards, we will find a control method suitable to solve an issue regarding the microgrid. In the long term goal, we may also model the microgrid with additional hydrogen system with a fuel cell and/or electrolyzer.

In the beginning, literature was read and surveyed to learn case studies of RES microgrid and control of RES microgrid.

Keywords: Microgrid, PV, solar energy, battery storage system, Maximum Power Point Tracking, Hydrogen Energy System, Fuel cell, power system, Power sharing, power-conversion, inverters, Energy Management System, DC Microgrid, Hybrid DC/AC microgrid

1 Introduction

There is increasing need for integration of renewable energy source (RES) and energy storage (RESS) into electrical power systems due to environmental and sustainability concerns. One of the ways this can be achieved is by adding RES and RESS into a microgrid. A micro-grid is defined as a combination of a source of electricity and a load, both of which have been localized. A MG is connected to main grid at the point of common coupling (PCC). It can function in two modes—either autonomously or in synchronization with the main grid. The operation of a MG is very similar to that of the main grid with the exception that it is meant for a smaller geographical area. It operates as a single controllable entity. [1]

Generally, microgrid is the composition of distributed generation(DG), loads, energy storage system(ESS), power electronic converters(PECs) and control devices; but the basis of microgrid is distributed resource (DR) that is the summation of DGs and ESS : $DR = DG + ESS$. DG refers to a small-scale power system that may be independent of the large electrical grid and are primarily located on the consumer side to meet their demands, whereas ESS stores energy in batteries, regenerative fuel cell(FC), and other device. The DR and DG technologies are a subset of distributed energy sources (DER) that is defined as the generation or production of electricity or heat on the load (or demand) end for local application. [2] Microgrids are being used to improve reliability and resilience of electrical grids, to manage the addition of distributed clean energy sources such as solar photovoltaic (PV) to reduce fossil fuel emissions, and to provide electricity in areas not served by centralised electrical structure. Microgrids, which are composed of distributed generation systems, energy storage, and loads, are being developed in various areas for both low and medium voltage ranges. [3]

Microgrids often include technologies like solar PV (which outputs DC power) or microturbines (high frequency AC power) that require power electronic interfaces like DC/AC

or DC/AC/DC converters to interface with the electrical system. [4] Some renewable energy sources such as PV, fuel cells, and batteries are DC. [3]

In the beginning, we do an initial study of designing and modelling in Simulink MATLAB a simple microgrid consisting of a PV array, a battery, load connected to a grid which is represented by an AC voltage source. As PV and battery is in DC, we need to do a DC-AC conversion to connect to the grid. The PV array, which is connected to a DC boost converter, is controlled by using Maximum Power Point Tracking (MPPT) control. If successful, we will then continue with adding hydrogen energy system such as fuel cell and electrolyzer into the microgrid.

First, we do a literary survey of types of microgrid, examples of microgrid control, and challenges with a microgrid - whether general or integrated with a RES.

2 Microgrid Architecture

According to system architecture, microgrids can be classified into three types: [3]

1. DC Microgrid

This type of microgrid is flexible due to its capacity of natural interfacing with DC-based DG, renewable energy source (RES), loads, and ESS along with the efficiency due to minimal power conversion. [2] Also, there is less power loss associated with DC-AC conversion when DC-RES (e.g PV and FC) is connected to a DC microgrid rather than to the typical AC grid. The saving of energy using DC microgrid is approximately 35%. There is no need for voltage impedance matching unlike in the AC microgrid. [5] A sample schematic of a DC microgrid is obtained from [6] in Figure 1:

2. AC microgrid

The most commonly applied conventional type of microgrid. Several types of DERS such as PV, wind turbines, and FCs are connected and merged into the large power network or existing utility grids. Due to laid down network, AC microgrid requires minimum modification and it brings out minimum alteration to the topology. This system is

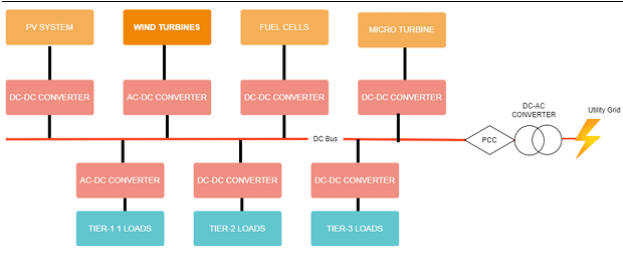


Figure 1: DC microgrid system layout. Image taken from [4]

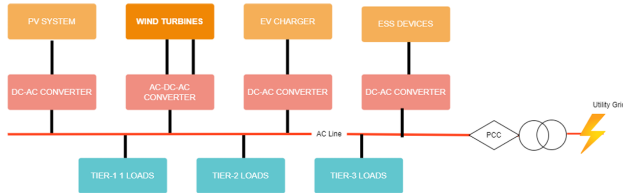


Figure 2: AC microgrid layout. Image taken from [6]

commonly merged with low and medium voltage levels due to its capability to inflate the distribution network with reduced transmission losses.

Though the interconnection of microgrid with conventional architecture is beneficial, it brings additional cons, such as system stability, power quality, reactive power deficiency, and DERs synchronization. [2] A sample schematic of AC microgrid system layout is obtained from [6] in Figure 2.

3. Hybrid AC/DC microgrid

Hybrid microgrid is the interconnection of both AC and DC microgrid(s). Though the network architecture of hybrid microgrid system is complex, it offers pros linked with both of the microgrid(s) such as flexibility, increased efficiency and reliability along with economics operation. The hybrid topology facilitates smooth interconnection with conventional grid due to AC microgrid architecture, whereas DC microgrid architecture helps in amalgamation of DC-based RES, DERs, ESSs, and loads with minimal alteration in the existing system. The hybrid microgrid topology drastically reduces the number of PECs required followed with the cutting down of unnecessary losses due to power conversion. [2]

The sample figure of an architecture of the hybrid AC/DC microgrid is obtained from [6] in Figure 3.

Due to the DC nature of PV and hydrogen fuel cell, we may consider either a DC microgrid or a hybrid DC-AC microgrid in our model in the future. The hybrid one considered is DC subgrid consisting of RES and DC ESS, and AC subgrid

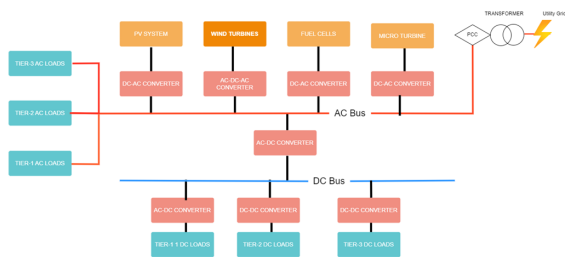


Figure 3: Hybrid AC/DC microgrid. Image taken from [4]

consisting of AC loads.

3 Challenges and Some Research Gaps of a Microgrid

There are various challenges associated with the operation of MG such as [1]:

1. There is an inherent risk of bi-directional power flow due to the presence of distributed generators at low voltage which could lead to faults.
2. With changes in the operating point, the output of the DG tends to deviate and these deviations for different DGs can add up and produce oscillations in the generated power.
3. The low inertia of the MG may produce severe frequency deviations and eventually lead to the loss of synchronization.
4. The intermittent nature of renewable energy system such as solar energy (and wind energy) poses a problem in stable micro-grid operation.

According to Khisa [6], the use of data collected from different components in the microgrid in addition to monitoring and forecasting of weather patterns will give the microgrid more control abilities over uncertainties. Research is still required on what will be required of DER to report to microgrid controllers in terms of measurement. He also observed several research gaps in microgrid research such as:

1. Identification of critical device information for control and modeling, including inverter transient response characteristics and operational limitations

Often microgrid controls operate on the assumptions of device behavioral characteristics, which opens a big gap when considering optimal control methods. If the devices connected to the system could be made to report their operational data, this information could be fed into the controller to make the system more adaptable.

2. Hierarchical, distributed and hybrid control strategies for devices, microgrids, and networked microgrids

There are many solutions for microgrid control. However, one should consider a solution that supports multiple control methods to meet each microgrid mission. The modularity of the controllers is one method of making controllers adaptable to scaling, different optimization strategies, etc.

4 Hierarchical Control

To fulfill the control aspects of the microgrid, three levels of control strategies are proposed. In [2], there are three levels (discussed specifically for an AC microgrid): primary control, secondary control and tertiary control. In [6] and [7], there are 4 levels: Primary control, secondary control, central control and global control. For brevity, the three layers in [2] will be discussed.

1. Primary Control (Bottom layer control)

The primary control scheme is directly connected to the microgrid and controls the fluctuations during the transition of the microgrid, that is, switching (or transition) from grid-connected to islanded mode. The primary

task of the p.c is among them to maintain stable voltage and frequency; and ensuring active and reactive power sharing between parallel operation DERs. This scheme is consisted of current control or/and voltage control loops of DERs. In this the voltage source converters (VSCs) could be utilised either as current-controlled or voltage-controlled voltage inverter.

2. Secondary Control

To regulate and minimise the deviations (such as voltage and frequency) in the steady-state parameters, secondary control scheme is proposed. This scheme utilises optimisation strategies at control methodology, and thus, requires communication channel configuration as bi-directional in nature. Further, this scheme is classified into decentralised control, which is used in large microgrids and central control (CC schemes), which is used in small microgrids.

3. Tertiary Control (Higher layer control)

The t.c looks after economic operation of the microgrid by adjusting the operation of the microgrid by adjusting the DERs power references. Although this scheme accelerates the economic operation of the microgrid, it is the slowest control scheme when compared with the other two former schemes. Optimum economic operation is achieved when all DERs are operated at the same marginal cost. Different optimisation algorithms such as particle swarm optimisation are engaged to ensure effective and accelerated economic operation.

5 Grid-connected Microgrid and Autonomous/Grid-Independent Microgrid

The duty of the grid-connected MGs as well as single DERs, whose power ranges are not considerable with respect to the main generation units of the grid, is to generate a certain amount of power and to maximise renewable energy generation. These entities are not responsible for frequency and voltage control/regulation. On the contrary, DERs in autonomus and grid-independent MGs have the main responsibility for frequency and voltage control. Therefore, DER's control objectives should be switched for these two operating modes between frequency and power control. These two general strategies, i.e frequency/voltage and power control strategies, are the main control goal of the microgrid.[5]

Initially, we are learning how to simulate a grid-connected microgrid, so we are focusing on the grid-connected microgrid goal which is to generate power and maximise renewable energy generation.

6 Control of DC microgrid

Several control methodologies such as distributed, decentralized, and centralized are proposed to achieve stable operation of DC microgrid. The following list depicts various aspects of the control strategies [2]:

1. stable voltage
2. reduced power loss
3. smooth transitions
4. power flow control among interconnected DC microgrids
5. load sharing

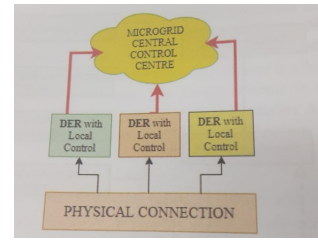


Figure 4: Framework of centralised control scheme. Image taken from [2]

6. power sharing
7. economic operation

The main components of the DC microgrid include battery storage and parallel DERs based on the converters. Each converter is regulated by a local controller having voltage, current, and droop control as variables. Some examples of the controller are such as the PV system is operated with a source-dependent controller named maximum power point tracking (MPPT) and battery (or ESS) system works with the state of charge controller. These local controller are centrally regulated for efficient management of energy and the control is defined as CC system.

On the basis of communication links, controllers could be classified into 3 types: (1) distributed control (2) Centralised system (3) Decentralised system [3]

1. Centralized control (CC)

- The DC microgrid having CC has a CC system along with digital communication networks which connects the generation sources and loads. (Figure 4) DERs in small-scale DC grids are controlled by CC having high bandwidth communication links(or networks) utilising the master and slave methodology, whereas hierarchical control is used in large-scale DC microgrids.

The hierachical control methodology comprises locally controlled inverters along with digital communication link (DCL)-based synchronisation control. Due to the introduction of substantial of substantial and considerable distinguished control between varied hierarchical levels, DC microgrid continues to operate even after the CC failure.

Though CC provides the worthy perfomrance, it cannot allow a single point of failure in the communication network, and therefore, requires an expensive redundancy in the complete communication network system. [2]

2. Decentralized control

- The decentralised control scheme is in Figure 5. This control method is based on the LC and some of the utilized methodologies include adaptive control of droop coefficient, power line signaling (PLS), and data bus signaling (DBS). The DBS control methodology takes the variation in DC bus voltage as an input to regulate several

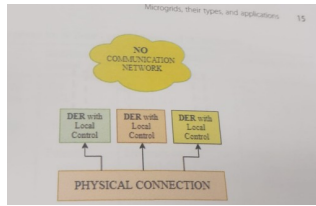


Figure 5: Framework of decentralised control scheme. Image taken from [2]

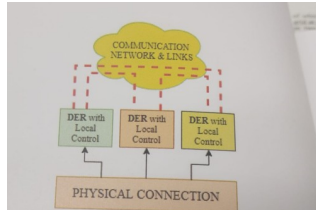


Figure 6: Framework of distributed control. Image taken from [2]

DERs, ESS, and parallel connected utility grid. The prime function of the DBS is the regulated and coordinated operation and control among several resources (DERs) such as PV, wind, and ESS, which is realised with the help of droop characteristics of current voltage plot. This control scheme is employed locally over DC bus voltage; thus communication links are absent making it simpler in implementation.

3. Distributed control

- There is no CC unit established and all local controllers communicate with each other through committed and customised DCLs. (Figure 6) Compared to CC scheme, this control could bear communication failure for a period of time. This control has better information consciousness and data realisation as compared to CC structure. However, the high level of complexity and stability margin are some of the challenges among the distributed control structure. [3]

7 Control of hybrid AC/DC microgrid

The hybrid microgrid is the combination of AC microgrid and DC microgrid. The AC microgrid is widely used and configured due to minimum alterations required in the existing infrastructure and utility grids. DC microgrid is gaining popularity due to its own advantages, e.g: no reactive power requirement/compensation, no synchronisation issue, increasing DC loads, minimum transmission losses, and economical to transmit power over long distances. Therefore the hybrid microgrid is an optimal configuration due to the pros from both AC and DC microgrids.

The hierarchical control schemes of this system has:

1. Primary control

- The main task of p.c is to ensure the quality of V/F and P/Q (V =Voltage, P = Active power, Q = reactive power) in permissible limits.

2. Secondary control

- The primary function of s.c is ensuring good power quality and minimization of V/F deviation occurring at primary level.

3. Tertiary control

- The key function of t.c include minimising the deviations occurred by second control, managing the demand, power exchange, and market pricing.

8 Nonlinear (More Advanced) Modelling and Control Issues

Microgrid (MG) controllers are typically designed using linearised models that are centered around the system's operating points for different control layers. However, in an MG that employs converter-based distributed that employs converter-based distributed energy sources (DERs), its main dynamic modes tend to vary stochastically due to changes in system parameters. Thus, a fixed linearised time-invariant model may not adequately capture the MG's behaviour, and its assumptions may not be valid under such circumstances. Furthermore, associated designed controllers using these simple models may not work well for certain control objectives (e.g harmonic mitigation) under system nonlinearities, uncertainties, and constraints. To address this issue, advanced modelling and control techniques have been proposed to enhance the accuracy and performance of MG controllers. Also, classical control techniques such as PI (proportional-integral) control design methods may not be effective for certain control objectives. [7] PI control has an inadequate response to disturbances, and limited adaptability to changing conditions and multiple objects. [8] Different advanced control synthesis methods such as nonlinear control, optimal control, robust control, and AI-based control have been proposed and applied to MG over the past decade. These techniques are better applied to handle the MG's nonlinear dynamics and system uncertainties and can offer superior performance compared to the classical controllers. [7]

This issue is important for consideration into designing a renewable energy-microgrid as renewable energy such as PV and wind is intermittent and is nonlinear in nature.

9 Solar-Battery Microgrid Literature

A solar generation based-microgrid has emerged as a viable substitute to fulfill energy demand while addressing environmental concerns [1]. Solar cell is the most abundant form of renewable energy. Its usage has increased considerably as it has become more feasible and convenient. Power electronics played a major role in this area. Electronic converters help in converting the DC output power of the PV arrays into AC power, which can then be used for supplying local loads or the grid. Batteries are used to store this power in case the load demand is less. [9] However, stochastic nature of solar irradiance presents operational challenges. [1]

The characteristic of the PV cell is nonlinear in nature. The presence of factors such as irradiance and temperature in the PV's output current equation affects the nature of the current/power and voltage curve. [5] (See Figure 7)

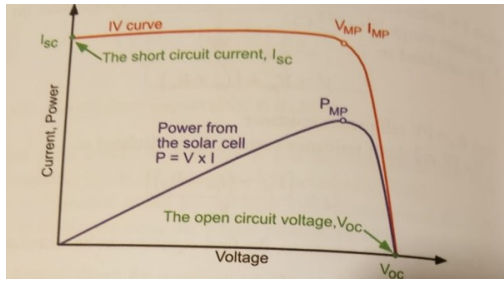


Figure 7: I-V curve of a solar cell. Image is taken from [5]

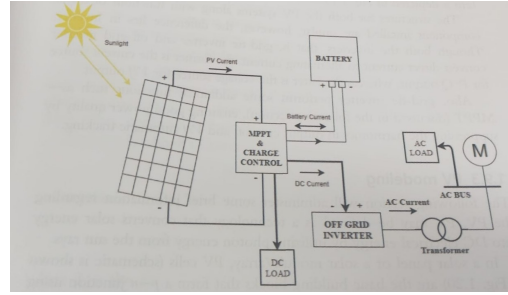


Figure 9: Independent PV system[2]

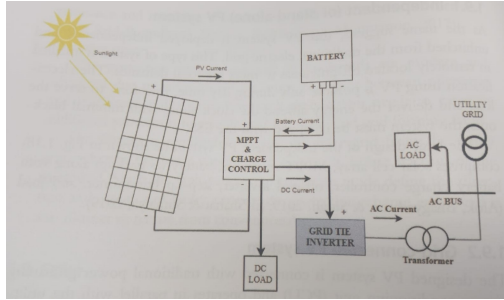


Figure 8: Grid-connected PV system [2]

A PV system microgrid has two types: Independent(stand alone system) and grid-connected. A sample independent PV system is in Figure 9 and a grid-connected PV system is in Figure 8 . An independent PV system is deployed independently of the grid. The electrification using PV is possible only during daytime, therefore, to serve the load and deliver the energy along with minimal blackouts, the system must be equipped with the ESS. Whereas, a grid-connected PV system is connected with traditional power grid using power conditioning unit (PCU) and operates in parallel with the utility grid. It injects power to the grid with the help of PCU that converts the PV DC output into AC power with grid's required level of voltage and power quality. A centralised type grid-connected PV system directly transfers power into the grid to consumers, but a distributed-type grid-connected PV system has the power directly transmitted to serve the load and fulfill the demand in the microgrid. Both PV systems have a solar cell array, MPPT controller, battery along with battery state controller, off-grid(for independent PV system)/grid-tie inverter (for grid-connected PV system). The off-grid inverter is the current source for P/Q output. The grid-tie inverter has several functions: voltage source for voltage/frequency (V/f) output, V/f automatic tracking, and ensuring good power quality by suppressing the harmonics in output current. [2]

Commonly in a solar-based microgrid, Maximum Power Point Tracking (MPPT) is used to track the maximum power point of the PV. PV is connected to a DC-DC (buck)-boost converter, in which the MPPT function can be implemented. [1,2,5,7,9,10] A battery energy system consists of a battery, a bidirectional dc-dc converter, and a current controller to regulate output current/power. The current controller can be considered as a PI controller. A BESS requires a bi-directional power converter to have the ability to be charged or discharged. The storage system has several

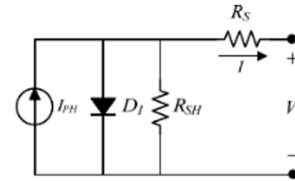


Figure 10: Equivalent circuit of a PV cell. Image extracted from "A Matlab/Simulink Based Photovoltaic Array Model Employing SimPowerSystems Toolbox by Massoud et.al

constraints such as minimum/maximum state of charge (SOC), minimum/maximum rate of charge/discharge, and state of health. [7]

9.1 Sample Components and System Modelling of Solar-Battery Microgrid

Laguana simulated a renewable microgrid model in Simulink using a PV array, battery, power converters, and load. [10] In the solar generation part of his model, the purpose of the model is to simulate a PV array by applying all the equations which describe its behaviour. The model also simulates the effect of the Maximum Power Point Tracking algorithm in a very simplified manner, and the DC/DC converter which injects power into the DC bus. The storage model part of the whole microgrid system has two main components: the battery and the bidirectional DC/DC converter which charge and discharge the battery at the required voltages.

The system modelling of the components are used as a reference. The images of the electrical circuit of PV cell and array (Figures 10 and 11) are taken from Laguana's thesis, which in turn is actually taken from other articles he used as a reference for his work. The only external outputs of this PV array block are: the solar irradiance G , the temperature of the cells T_c , and the reference DC voltage in the DC bus $E_{DC}^* = 800V$

Solar Generation: PV cell A solar cell can be described as a photo-generated current source, a diode and two resistances (one shunt resistance and one resistance in series . (Figure 10)

Where I_{ph} is the photo-generated current, which depends on the irradiance and the temperature of the cell; R_{sh} and R_s are the shunt and series resistances; I_d (not shown in the figure) is the diode current, and I_{sh} are the losses in the shunt resistance. I and V are the output current and voltage

of the cell.

$$I = I_{ph} - I_d - I_{sh} \quad (3)$$

The equation for the photo-generated current is:

$$I_{ph} = \frac{G}{G_{ref}} [I_{sc} + K_{ti}(T_c - T_{c,ref})] \quad (4)$$

Where $G_{ref} = 1000 \text{ W/m}^2$ is the reference irradiance (standard conditions), $T_{c,ref} = 25^\circ\text{C}$ is the reference cell temperature (standard conditions), I_{sc} is the short-circuit current at $T_{c,ref}$ and G_{ref} (provided by the manufacturer), and K_{ti} is the cell short-circuit current temperature coefficient.

The equation for the direct current in the diode is:

$$I_d = I_o \left(\exp \left[\frac{V + IR_s}{\eta V_t} \right] - 1 \right) \quad (5)$$

$$V_t = \frac{\sigma T_c}{e} \quad (6)$$

$$I_o = I_{oref} \left(\frac{T_c}{T_{c,ref}} \right)^3 \exp \left[e E_g \left(\frac{1}{T_{c,ref}} - \frac{1}{T_c} \right) \right] \quad (7)$$

$$I_{oref} = \frac{I_{sc}}{\exp \left[\frac{V_{oc}}{N_s V_t} \right] - 1} \quad (8)$$

Where I_o is the reverse saturation current of the diode, $I_{o,ref}$ is the reference reverse saturation current (standard conditions), V_t is the equivalent voltage for temperature, $\sigma = 1.380648 \cdot 10^{-23} \text{ J/K}$ is the Boltzmann constant, $e = 1.602176 \cdot 10^{-19} \text{ C}$ is the charge of one electron, $\eta \in [1, 2]$ is the ideality factor of the diode, E_g is the Gap Energy, V_{oc} is the open-circuit voltage at $T_{c,ref}$ and G_{ref} , and N_s is the number of series connected cells in a single PV module (the full array consists in series and parallel connected modules, it's important to remark that it has been considered in the model that there are not parallel connected cells in a PV module, otherwise the equations would be slightly different. The equation for the losses in the parallel branch is:

$$I_{sh} = \frac{V + IR_s}{R_{sh}} \quad (9)$$

PV array The PV Array is a series-parallel association of individual modules (which are also a series association of individual cells) This association creates a solar generation which depends on the PV module used (assuming that all the modules are the same), on the parallel-series connected modules, on the irradiance and on the temperature of operation. (Figure 11) The behavior of the full array is described by equation 10:

$$I = m_p I_{ph} - m_p I_o \left(\exp \left[\frac{\frac{V}{m_s} + \frac{IR_s}{m_p}}{\eta V_t} \right] - 1 \right) - \frac{\left(\frac{m_p}{m_s} V + IR_s \right)}{R_{sh}} \quad (10)$$

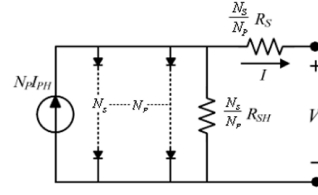


Figure 11: equivalent circuit of a PV Array. Image extracted from "A Matlab/Simulink-Based Photovoltaic Array Model Employing SimPowerSystems Toolbox"

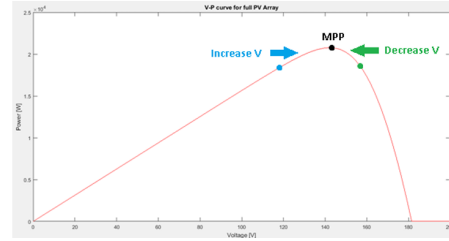


Figure 12: Hill Climbing concept in a V controlled P&O MPPT algorithm. From [10]

Where (m_p, m_s) are, respectively, the parallel connected modules and the series connected modules in the array.

MPPT and DC/DC converter The maximum power point tracking is a technique used with photovoltaic arrays to maximize the power extraction regardless of the irradiance and temperature conditions. In this case, the MPPT algorithm tracks the maximum power point and injects the maximum power to the DC bus through a DC/DC boost converter which converts the voltage from $V_{mp}(G, T_c)$ to $E_{DC}^* = 800\text{V}$. The MPPT algorithm computes the required duty cycle D for the boost converter in order to guarantee maximum power. The Perturbe and Observe (Hill-Climbing) MPPT algorithm is used. This algorithm compares the produced power in the instant (t) with the produced power in the previous instant ($t - \Delta t$) and changes the duty cycle of the converter in order to increase/decrease the value of the current I_{PV} as needed. The power supplied by the PV Array after the MPPT is always oscillating around the maximum power (Hill-Climbing concept), shown in Figure 12. The MPPT and various MPPT algorithms will be discussed in the later section of PBL1 Progress Report.

DC/DC Boost converter The boost DC/DC converter is a power electronics converter which gives a higher output DC voltage from a lower input DC voltage. Their components are: an inductance in series with the input voltage source, a switch in a parallel branch, a diode in series with the inductance, and a capacitor in a parallel branch. In this case, the switch (transistor in commutation) is controlled with a PWM technique which can be described by the duty cycle (D). The duty cycle is computed by the MPPT every Δt (as described previously).

In the simplified MPPT + DC/DC Boost converter subsystem in [10], the MPPT is substituted for a function which injects the maximum power to the average DC/DC converter which injects this power into the bus at $E_{DC}^* = 800\text{V}$. The

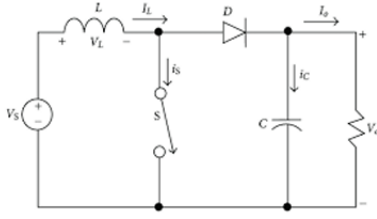


Figure 13: DC/DC Boost converter. Image extracted from “Electrónica. Grau en Enginyeria de Tecnologies Industrials, ETSEIB – Emili Lupon

results show that the system takes some time to reach the maximum power. When the power gets its maximum, the value of the DC current stabilises. The circuit of DC/DC boost converter is in Figure 14.

Battery Storage System The function of this model is to simulate how the storage system of the microgrid works. It contains two main components: the battery and the bidirectional DC/DC converter which charge and discharge the battery at the required voltages.

Battery used to store the energy which is not used in certain moments of operation (When the demanded power is lower than the generated power). This energy can be supplied to the system when the demand has a peak and the PV solar generation is not enough to cover it. The rated capacity of the battery is its energy capacity under nominal conditions and it's provided by the manufacturer. The State of charge is the level of available energy in the battery in relation with its capacity.

DC/DC bidirectional converter This converter has two main functions: o Inject power from the DC bus (800V) to the battery (around 200V): Buck o Inject power from the battery (around 200V) to the DC bus (800V): Boost The required converter in this case is a bidirectional Buck-Boost converter.

9.2 Sample Energy Management of Solar-Battery System

In Gaurav et. al's paper[9], a PV array output is boosted to the required value of the load. The battery is connected via a dc-dc bidirectional converter. The dc-dc converter uses IGBT for its fast switching speed and low output impedance. This paper proposes the integration of a non-conventional source of energy such as PV array with a local load and battery system along with various modes of operation. These modes depend upon the PV output and battery state of charge (SOC) conditions. The block diagram of the proposed system is in Figure 14.

1. The Solar PV Module used in this paper acts as an input source for charging the battery as well as supplying to the AC Load during normal conditions. The PV Module operates at constant irradiation level of 1000 Wm⁻² and ambient temperature of 25 °C. At optimum conditions produces an output of 23 V. The module is connected to the AC Load through a DC-DC Boost Converter and 1-Phase Inverter. It is connected to the

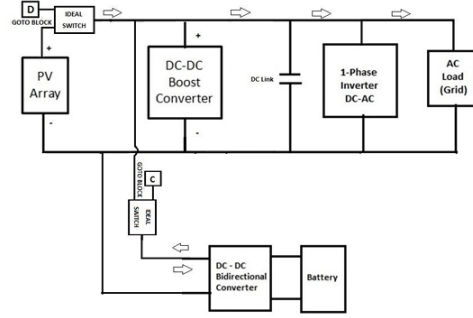


Figure 14: Block diagram of Gaurav. From [9]

Battery through a DC-DC Bidirectional Buck-Boost Converter.

2. In PV systems, generally, dc-dc converters are used either to step up/down the PV voltage. They can also be used to extract maximum power through MPPT (maximum power point tracking). During continuous conduction mode of operation this converter has two intervals of boost operation.
3. A non-isolated DC-DC Bidirectional Buck-Boost converter is used in this paper to boost PV output to battery voltage requirement during charging conditions and in discharging conditions reduce battery output voltage to desired value to supply the load.

For the Buck Converter operating in Continuous mode, the duty cycle is given by

$$\frac{V_{out}}{V_{inp}} = 1/(1 - D_{boost}) \quad (11)$$

The current at the continuous-discontinuous boundary is given by

$$I = \frac{DTs(V_{out} - V_{inp})}{2L} \quad (12)$$

$$L = \frac{(2 \times 10^{-5})(0.209)(110 - 23)}{2 \times 30} \quad (13)$$

4. Controller Logic

The control logic uses relational operator to check when the PV insolation levels or the load current change. It can be seen that the IGBT switches of the Bi-directional converter are controlled using multiport switches. Boost mode IGBT is controlled using [A] Goto block and buck mode IGBT is controlled using [B] Goto block. It can be seen that the Battery module is connected to the main system using ideal switch controlled by [C] Goto block. Also, the PV array is connected to the system using ideal switch controlled by [D] Goto block. As soon as there is a change in the parameters the control logic generates pulses in terms of 1 or 0 for each Goto block to change the modes of operation of the system.

9.3 DC-subgrid in a Hybrid Microgrid

In 2021, Nguyen [3] proposed a hybrid power system combining a conventional AC main grid with a DC sub-grid. Each prosumer connects to the DC sub-grid through a

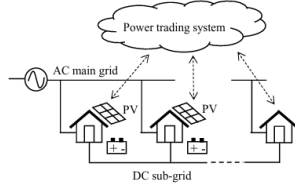


Figure 15: The hybrid electrical power system with a DC sub-grid. Image taken from [3]

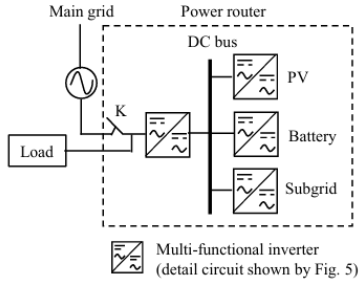


Figure 16: The structure of the power router [3]

device called a power router, which uses half-bridge inverters to manage power flow. Unlike an AC sub-grid, the DC sub-grid does not require synchronization between nodes, allowing independent control of each prosumer without shared information. The hybrid electrical power system with a DC sub-grid schematic is in Figure 15 and the structure of the power router is in Figure 16.

The power router regulates power exchange between the main grid, DC sub-grid, PV generators, batteries, and loads. It maintains a constant DC bus voltage by ensuring that the total input and output power is balanced. For example, when a PV system generates excess power and raises the DC bus voltage, the inverter transfers the surplus power to the AC grid to stabilize the voltage.

The system can operate both grid-connected and independently. If the main grid fails or operates outside standard voltage or frequency limits, the power router detects the fault and either continues operation using fault ride-through (FRT) control or disconnects from the grid. In such cases, the system can function as an islanded DC microgrid. The power router also enables stand-alone operation, allowing PV or batteries to supply loads without connection to the main or sub-grid.

Additionally, the system supports power trading between prosumers based on supply and demand. Each prosumer communicates with a trading platform, and the power router adjusts power transfers through the DC sub-grid according to transaction results. Throughout operation, the router's controller continuously monitors inverter currents and voltages, shutting down the system if abnormal conditions are detected.

For our simple microgrid, we may adopt a similar model to that in Figure 16.

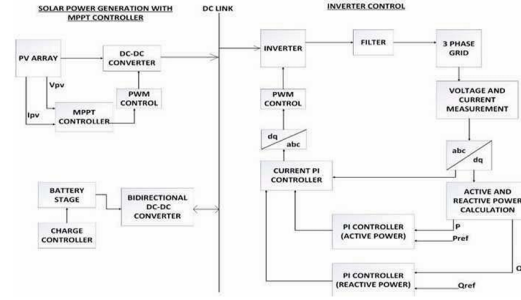


Figure 17: Schematic of a solar-based MG operating in PQ mode [1]

9.4 PQ Control

Shrivastava et. al [1] seek to implement PQ-control of a solar-battery microgrid operating in a grid-connected mode. The schematic of the system is in Figure (?). The PV array is equipped with MPPT mechanism, and the boost converter is equipped for realising the MPPT P&O algorithm. The system has been tested for two cases in Simulink MATLAB: constant load and variable irradiance, and variable load and variable irradiance. PQ control is used for optimum MG operation. In this mode, the d-axis and q-axis reference currents are determined based on the reference active power and reactive power. The PV array is equipped with MPPT controller using P&O algorithm. The active and reactive power being fed from solar generator is compared with reference values. A PI controller error is used for processing of error. The control circuitry ensures inverter output in accordance with the reference values. Thus PQ control of micro-grid can be achieved. The inverter has been modeled for three phase application. The DC link capacitor acts as the input for the inverter. The pulse width modulation (PWM) has been taken care of in the control structure of the inverter. Control action takes place in a synchronous reference frame. Park's transformation has been used to convert the three-phase voltages and currents from abc to rotating d-q reference frame. The DC-DC Boost converter performs the function of stepping up the output voltage of the PV array to the required level.

$$Power_{active} = V_d * I_d \quad (14)$$

$$Power_{reactive} = -V_q * I_q \quad (15)$$

Using Eq. (14) and Eq. (15), the reference currents can be calculated as:

$$i_{d,ref} = Power_{active,ref} / V_d \quad (16)$$

For the proper implementation of the P-Q control method, the need for battery storage and a charge controller is inevitable to make sure there is a balance of power. When the solar power is greater than demand, the surplus power is absorbed by the battery system through a bi-directional converter. Conversely, when there is a deficit of power from the PV array, the battery storage system fulfills the demand. The

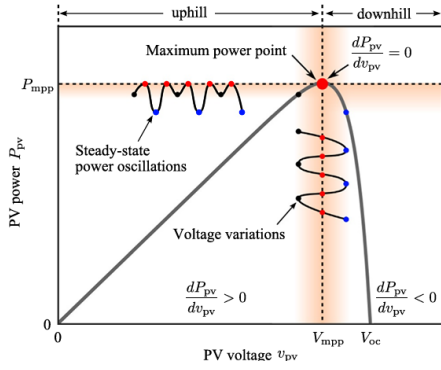


Figure 18: Power-voltage (P - V) characteristic of PV modules with MPPT control [11]

results indicate that adequate PQ control has been achieved for variable load and irradiance and the battery system works efficiently.

10 Control of PV: Maximum Power Point Tracking (MPPT) control

Unlike conventional resources, PV's generation and control of power cannot be achieved with the variation in the fuel inflow rate or amount of energy transferred to the generator; the additional controller (MPPT) is used to guide and track the maximum power point (MPP) of the PV system. [2] To maximize the power production (i.e., optimize the energy capture), MPPT control is necessary for PV systems. [11] The MPP is highly affected due to nonlinearity in the relation(s) between voltage-current and voltage-power because of varied insolation level, weather condition and temperature. Thus, to overcome these effects, MPPT techniques are exerted based on various algorithms proposed to extract the maximum possible power from the PV module. [2] The MPPT action can be incorporated into the DC-DC boost converter. [1]

Actually, most of the reported MPPT control schemes are realized in accordance to the P - V characteristic of PV modules, which is a hill-like curve. This is shown in figure 18. More specifically, on the top of the hill, the PV modules produce the maximum power, called the MPP. On the left side of the MPP, the output power has a nearly linear relationship with the PV operating voltage; while above the MPP voltage, the power drops with the increase of the PV voltage. Ideally, MPP control should maintain the PV module operating point at the MPP to be robust and act quickly in the face of environmental variations (e.g., passing clouds inducing a sudden drop in the solar irradiance level). [11]

Two main MPPT algorithms, the previously mentioned P&O and also Incremental Conductance (INC) are briefly explained [5]:

Perturbation & Observation Algorithm The flowchart of the algorithm can be seen in Figure 19. In the P&O MPPT algorithm, the output of the PV module is first perturbed and voltage, $V(t)$ is made unsettled. Accordingly, power, $P(t)$ is calculated and compared with $P(t-1)$ followed by comparison between the last measured voltage, $V(t-1)$ and

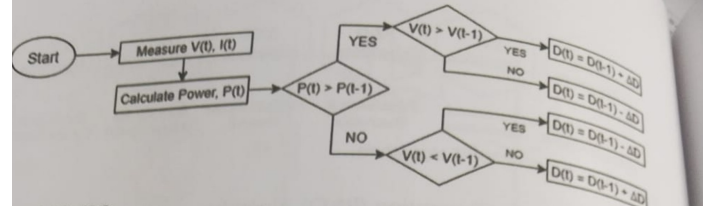


Figure 19: Perturb and Observe algorithm flowchart [5]

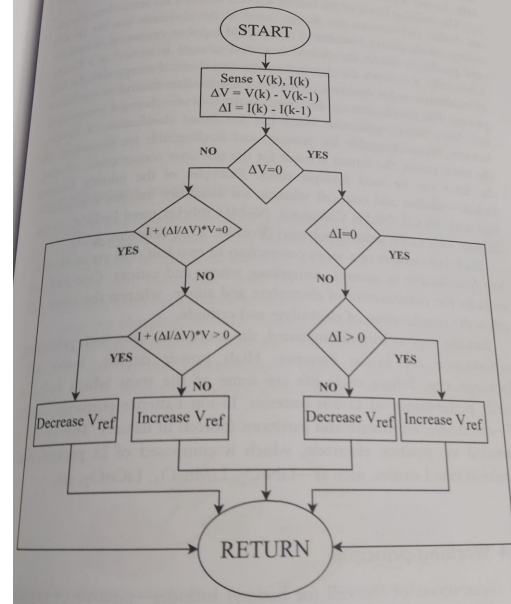


Figure 20: Incremental Conductance flowchart [5]

present voltage, $V(t)$. If the measured power, $P(t)$ is greater than $P(t-1)$, along with higher $V(t)$ as compared to $V(t-1)$, then the Duty cycle (D) is incremented by a constant factor ΔD , else D is decremented by the same factor, that is, ΔD .

Similarly, if $P(t)$ is found lesser than $P(t-1)$ followed by comparison between $V(t)$ and $V(t-1)$, the Duty cycle (D) is incremented (by a factor ΔD) if $V(t)$ is greater than $V(t-1)$, else decremented by a factor ΔD . The variations are maintained and controlled by a constant value ΔD , therefore P&O algorithm fails to cater the system during the rapid change in irradiance level and abnormal variation(s) in the atmospheric condition.

Incremental Conductance(INC) algorithm The flowchart of INC algorithm is in Figure 20. INC is better than P&O in that this can cater the system with rapid change in parameters. The principle of INC algorithm is the comparison of PV module INC with instantaneous conductance and the following characteristics of PV's derivative power: (1) zero value of dP/dV at MPP, (2) positive value of dP/dV at MPP's left, and (3) negative value of dP/dV on MPP's right, forms the basis and monitors an algorithm.

Mathematically, the former relations can be denoted as:

At MPP,

$$\frac{dP}{dV} = 0 \quad (1)$$

At MPP's Left

$$\frac{dP}{dV} > 0 \quad (2)$$

At MPP's Right

$$\frac{dP}{dV} < 0 \quad (3)$$

The principle of INC can be mathematically represented as:

$$\frac{dP}{dV} = \frac{d(IV)}{dV}$$

$$\frac{d(IV)}{dV} = I + V \left(\frac{dI}{dV} \right) \cong I + V \left(\frac{\Delta I}{\Delta V} \right)$$

For attaining the condition of MPP,

$$\frac{dP}{dV} = 0$$

Or,

$$I + V \left(\frac{\Delta I}{\Delta V} \right) = 0$$

$$\frac{\Delta I}{\Delta V} = -\frac{I}{V}, @MPP$$

$$\frac{\Delta I}{\Delta V} > -\frac{I}{V}, @MPP's Left$$

$$\frac{\Delta I}{\Delta V} < -\frac{I}{V}, @MPP's Right$$

where

$\frac{\Delta I}{\Delta V}$ = incremental conductance

$\frac{I}{V}$ = instantaneous conductance

Depending on the result, the V_{ref} (previous MPP) voltage is increased or decreased and gets replaced by the new voltage at MPP. The condition is maintained at MPP and V_{ref} until a change is occurred in the parameters such as ΔI , irradiance level, abnormal and sudden change in atmospheric conditions.

11 Hydrogen Microgrid and Control Literature

Hydrogen energy system is a promising energy system alternative due to decrease fossil fuel emissions and being sustainable. Hydrogen has a clean combustion, producing only water as its product. Hydrogen energy such as fuel cells, as mentioned before, is DC. Hydrogen storage gains preference over BESS owing to its superior energy density, extended lifespan, rapid refuelling/recharging, scalability, and potential for reduced environmental impact. Fuel cells (FCs) have recently garnered considerable interest as a clean power generation technology. Integrating FCs into HMG results in pollution-free and reasonable energy. The technology harnesses a chemical reaction to convert hydrogen and oxygen into electricity, generating heat and pure water as byproducts.[12]

Alzahrani et al. reviewed a hydrogen-based hybrid microgrid (HMG) integrating solar (PV) and wind (WT) energy with both AC and DC sub-grids, connected through an AC/DC power flow regulator. The system includes hydrogen storage (HSS), battery energy storage (ESS), and loads on

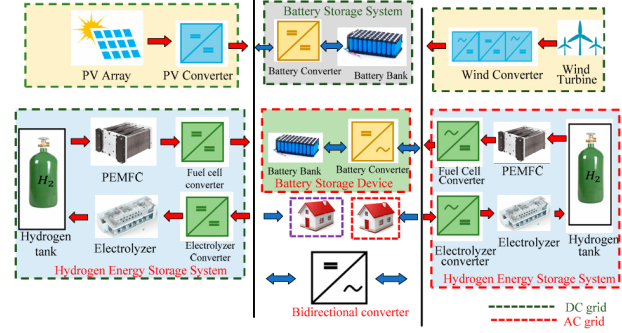


Figure 21: Multi energy storage system in hydrogen-hybrid micro-grid [13]

both sub-grids. (Figure 21) The hydrogen storage system acts as long-term energy storage: excess renewable energy powers an electrolyzer to produce hydrogen, which is stored in tanks. When needed, a fuel cell converts the stored hydrogen back into electricity at the DC bus. PV panels connect to the DC bus via a DC/DC boost converter, while wind energy is supplied through a permanent magnet synchronous generator connected to the AC bus. Bidirectional converters manage power exchange between the AC and DC sub-grids, ensuring stable operation. To maintain power quality, voltage stability, and power balance, the system uses a bidirectional AC/DC converter (BAC) with combined PI control. The BAC regulates real and reactive power by monitoring the DC link voltage and enabling controlled bidirectional power flow. The proposed control strategy significantly reduces reactive power (by about 90.3% and 89.4% in linear/ nonlinear loads), ensuring stable, balanced, and reliable microgrid operation. [13]

In [13], the HSS are modeled as the following. The modelling is discussed in this report due to its simplicity, compared to the more detailed and complicated modelling equations in [15]. At time t , the fuel cell's acquired energy E_{FC} is expressed in (17),

$$E_{FC}(t) = 37.8 * H_{2cons}(t) * \eta_{fc} \quad (17)$$

The electrochemical electrolyzer's hydrogen output H_{2form} is represented in (18),

$$H_{2form}(t) = \eta_{ele} \frac{P_{ele}(t)}{37.8} \quad (18)$$

with η_{fc} Fuel cell efficiency, H_{2cons} Consumed hydrogen level, E_{FC} Fuel cell's acquired energy, H_{2form} Electrochemical electrolyzer's hydrogen output, P_{ele} Input power from renewables, η_{ele} Efficiency of electrolyzer, and H_{2t} Hydrogen within the tank.

The hydrogen within the tank H_{2t} is replenished in accordance with the load needs when the renewable energy generated by the hybrid PV/WT cannot fulfil the demand.

$$\begin{aligned} H_{2t}(t) &= H_{2t}(t-1) + H_{2form} \\ H_{2t}(t) &= H_{2t}(t-1) - H_{2cons} \end{aligned} \quad (19)$$

Finding the right amount of hydrogen tanks, FC output power, and total electrolyzers will help to reduce the leveled cost of energy, which will reduce the overall cost. The electrolyzer separates water to highly pure oxygen and hydrogen by an electrochemical reaction. Figure 22 a depicts the electrolyzer's characteristic curve as a consequence of current and operational temperature. The relation between reference current and voltage is described by the following expression:

$$V_{EL} = E_n(T_{EL}) + E_{act}(I_{EL}, T_{EL}) + E_{ohm}(I_{EL}, T_{EL}) \quad (20)$$

The Nernst equation is the primary relationship utilized to describe the electrical activity of the electrolyzer or FC:

$$E_n = E_0 + \frac{RT_{FC}}{2F} \cdot \ln\left(\frac{P_{H_2}^{0.5} P_{O_2}}{P_{H_2O}}\right) \quad (21)$$

The fuel cell's characteristic curve at a 50 °C operating temperature is shown in Figure 22 b. By using a heat balance of the generator, the operating temperature of the fuel cell is calculated.

Research gap in hybrid MG (HMG) with HSS: Gupta et. al [12] observed in their literature survey that existing studies tend to overlook critical aspects such as frequency and voltage regulation, overload analysis of both AC and MG, and power sharing within HMG that incorporate RERs, BESS, hydrogen storage, and FCs. Instead, their primary focus remains on optimizing operation, sizing, placement, and cost analysis for HMGs. Even the literature that addresses these issues does so only within either DC-MG or AC-MG, not in a comprehensive manner for HMG. The literature review highlights a significant gap in the intelligent control structures for both grid-connected mode and islanded mode in HMGs. Thus, they proposed a novel intelligent scaled droop control structure (SCDS) for HMGs. This structure optimises grid operation by efficiently managing power balance and maximizing the utilization of generation capacity. It enhances grid capacity through smart load management and resilience during islanded operations, ensuring a reliable and sustainable power supply for smart cities and advancing future energy systems. The HMG architecture involves linking the AC-microgrid and DC-microgrid through bidirectional interconnection converters (ICC). This HMG combines AC sources like wind-DFIG with DC sources such as solar PV and solid oxide fuel cell (SOFC), supported by battery energy storage systems (BESS) and hydrogen storage units (HSU). The HSU can generate and store hydrogen during RER surplus. This stored hydrogen can be further employed for production of electrical power along with numerous other applications. To analyse the performance of smart city HMG, SDSCS has been proposed and analysed with real-time digital simulator (OPAL-RT 5600) to authenticate the results obtained with MATLAB. The proposed SDSCS efficiently manages HMG variables and demonstrates effectiveness in regulating frequency and voltage, even under excessive overload conditions.

Guo et. al [14] proposed a PV-hydrogen microgrid energy management using a hierarchical economic model predictive

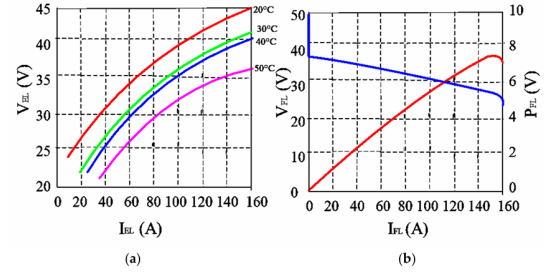


Figure 22: a Electrolytic characteristics b Fuel cell characteristics. From [13]

control framework (HEMC). the PHS microgrid system is composed of a PV energy source, a battery bank, a proton electrolyte membrane fuel cell (PEMFC), an electrolyzer, a hydrogen storage tank, an energy consumer and a connection to an electric utility (EU). In order to overcome the solar radiation uncertainties and maintain the system stable operation, a HEMPC framework is pro-posed, which includes two levels: a long-term EMPC and a short-term EMPC. The long-term EMPC is designed to maintain the PHS system's energy balance on a long-time horizon and achieve power load shifting with the aid of battery bank and the electrolyzer, while the short-term EMPC tracks reference signals from the long-term EMPC and re-optimizing the power demand tracking and the charging and discharging cycle of the hydrogen tank in real-time. To further verify the efficacy of the proposed HEMPC, a typical rule-based control (RBC) integrated with a MPC at the lower level is introduced for comparison purpose. The simulation results along with qualitative and quantitative analysis have demonstrated that compared with RBC-MPC, the proposed HEMPC is effective in managing the equipment power output, realizing dynamic synergy and enhancing the economic performance.

As [14] discusses about PV-hydrogen microgrid, we may use the modelling of the hydrogen components of the electrolyzer and PEMFC as a reference. The system modelling equations are not discussed here due to being complicated and long.

A relatively simple case of fuel cell electrical model is discussed in a design of a hybrid AC-DC microgrid consisting of wind, PV, biomass, hydro etc. The modelling of a PEM fuel cell equivalent circuit as as the following in Figure 23. When there is existence of contact of contact of different material then there is a charge transfer. It takes some time to response when there is abrupt change in the current due to double layer charge form voltage. Capacitance C functions as electrode with slack properties in the voltage double layer. [15]

The voltage across capacitance C is in eq. 22,

$$V_c = \left(1 - \frac{dv_c}{dt}\right) (R_{act} + R_{conc}) \quad (22)$$

The output voltage of the fuel cell V_{FC} , where

$$V_{FC} = E - V_c - V_{act} - V_{ohmic} \quad (23)$$

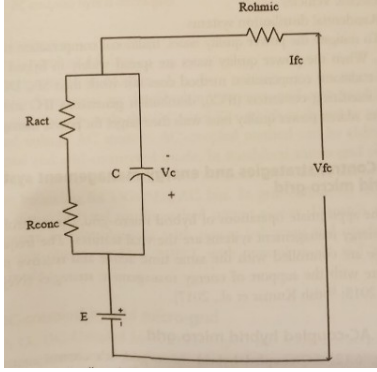


Figure 23: PEM fuel cell equivalent circuit

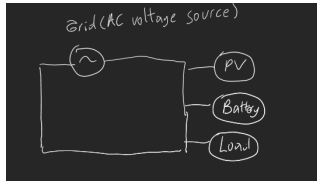


Figure 24: Initial microgrid concept

where R_{ohmic} , R_{act} , and R_{conc} are the ohmic, activation and concentration resistance.

12 Methodology: PV-Battery Microgrid

After the literature review, we begin our initial study of a simple grid-connected RES PV-battery microgrid consisting of a PV array, battery (Li-ion), load (series RLC branch), and grid represented as an AC voltage source simulated in MATLAB Simulink. As of now, we have only done the modelling and not a specific control problem yet. The PV array is connected to a DC-DC boost converter. To track the maximum power point of the PV array, MPPT P&O algorithm function is implemented in the boost converter. The DC battery is connected to the microgrid by a bidirectional converter to allow charge and discharge process of the battery. As the grid is AC, we also need a DC-AC converter. To track the dynamics of the microgrid, we simulate in seconds. The initial concept microgrid is in Figure 24.

To first learn a sample simulation of PV-Battery microgrid, I adapted a model from YouTube tutorial videos by Simscape Electrical. The DC part of the microgrid (PV-boost converter equipped by MPPT-battery) is adapted from [16] and the grid-connected part is adapted from [17]. The parameters of the PV array, capacitors, RLC load and battery are made the same as from [16]. I connected the MPPT-boost converter-battery structure to the grid using parameters of grid in [17]. The time sampling rate is set in discrete time $1 \text{ e-}6 \text{ s}$. The Simulink model of the whole microgrid is in Figure 25.

13 Simulink Components

13.1 PV array

The PV array we choose has 2 parallel strings and 9 series-connected modules per string. The module data is from Japan Solar Factory JB250P-60. The inputs to the PV array are sun irradiance in W/m^2 , and cell temperature in degrees Celcius

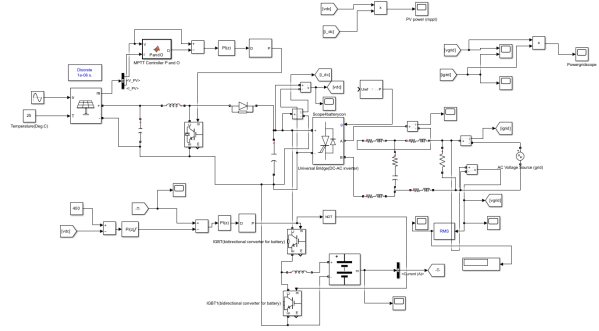


Figure 25: Simulink model of the PV-battery microgrid

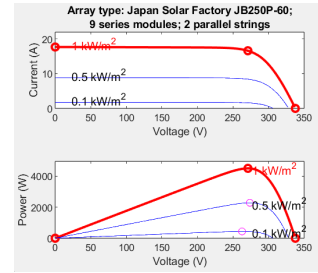


Figure 26: I-V curve and P-V curve of the PV array obtained under standard irradiance conditions in Simulink

(standard is 25°C . Under standard irradiance conditions 1000 W/m^2 and temperature 25°C , the PV array produces the I-V and P-V curve. The I-V curve of the PV array generated by Simulink is in Figure 26.

13.2 MPPT P&O

The output of the PV array is connected to a current and data bus, which in turned are connected to the MPPT P&O algorithm function. The MPPT algorithm is also implemented by the boost converter. (Figure 27) To better understand the MPPT function, we also substitute the irradiance input of the PV array by different function blocks (step, ramp, or sine functions).

The MPPT Peturb and Observe algorithm is represent by the MPPT controller P and O function block. Inside the function block, the MATLAB function code is shown below. The MPPT controller tracks changes in the duty cycle D and power P of the PV array. This will then form a feedback mechanism with the booster converter. The code is from

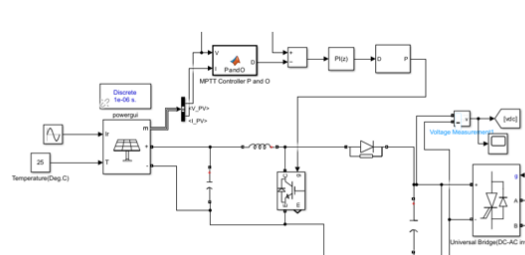


Figure 27: MPPT block of the Simulink model


```

[16]. % is MATLAB comment.
function D = PandO(V, I)
% MPPT controller based on the P & O algorithm.
% D output
% = Reference for DC link voltage (Vdc_ref)
% Enabled input = 1 to enable the MPPT controller
% V input = PV array terminal voltage (V)
% I input = PV array current (A)
% Param input:
Dinit = 260; % Initial value for Vdc_ref
Dmax = 290; % Maximum value for Vdc_ref
Dmin = 230; % Minimum value for Vdc_ref
deltaD = 0.03; % Increment value used to
                % increase/decrease Vdc_ref
persistent Vold Pold Dold;
P = V * I;
dataType = 'double';
if isempty(Vold)
    Vold = 0;
    Pold = 0;
    Dold = Dinit;
end
dV = V - Vold;
dP = P - Pold;
if dP ~= 0
    if dP < 0
        if dV < 0
            D = Dold + deltaD;
        else
            D = Dold - deltaD;
        end
    else
        if dV < 0
            D = Dold - deltaD;
        else
            D = Dold + deltaD;
        end
    end
end
else
    D = Dold;
end

if D >= Dmax || D <= Dmin
    D = Dold;
end
Dold = D;
Vold = V;
Pold = P;
end

```

13.3 Booster converter

The booster is equipped with MPPT algorithm. In the model, the output of the MPPT controller goes through a PI controller and fed into the booster. The booster then does a feedback loop and delivers the measured voltage and current to the PV array, and tracks the maximum power point of the PV array.

To model different MPPT responses, we subject different function shape blocks such as the step function, sine function

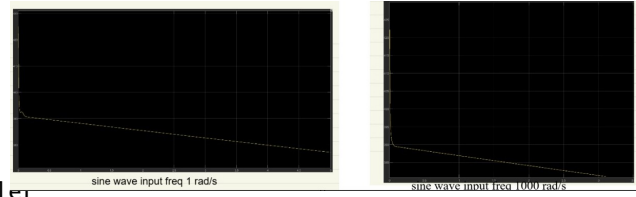


Figure 28: Battery-connected microgrid. Left image: MPPT voltage response when the solar irradiance is sine wave with frequency 1 rad/s and Right image: MPPT voltage response when the solar irradiance is sine wave with freq.1000 rad/s

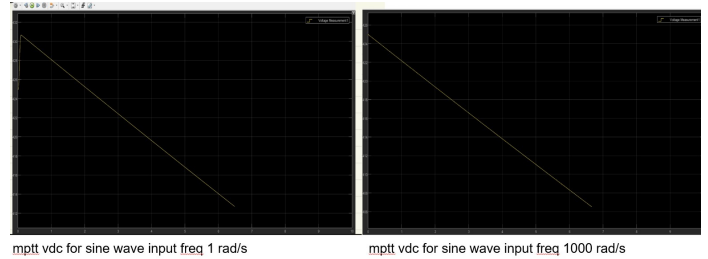


Figure 29: Microgrid not connected to the battery. Left image: MPPT voltage response when the solar irradiance is sine wave with frequency 1 rad/s and Right image: MPPT voltage response when the solar irradiance is sine wave with freq.1000 rad/s. Graph obtained from Simulink

and ramp function as the irradiance input to the microgrid twice: either battery connected or not connected to the microgrid. For brevity and ease of comparison purposes, only sine wave with frequency 1 rad/s and 1000 rad/s will be discussed. The output DC voltage of the MPPT subsystem generated by the sine irradiance is in Figures 28 and 29, for the battery-connected PV microgrid and the microgrid without the battery, respectively. As can be seen from the figures, specifically for sine wave solar irradiance in both battery-connected and battery-isolated microgrid, the maximum voltage of DC MPPT-boost converter voltage output V_{DC} is lower when the battery is connected to the microgrid. As the frequency increases, V_{DC} has less ripples and fluctuations. The microgrid is more stable when battery is connected to the system.

13.4 Battery Storage unit and bidirectional converter

Bidirectional converter For simplicity, the BCs is modelled by two IGBT blocks. The output voltage of the PV array is controlled and measured against the reference voltage 400 V through a PI controller. This feedback is then fed into the bidirectional converters, which connect to the battery. This will form the charging process of the battery.

Battery Storage Unit In this simple model, we use a Li-ion battery available in Simulink. The initial state of charge is 50%, and the capacity is 64 Ah. [15]

The battery is operated with a NOT logic, for either charging or discharging. From the battery, we measure the output current, which is then sent to a feedback PI control loop of the battery to compare with the reference voltage value of 400 V.

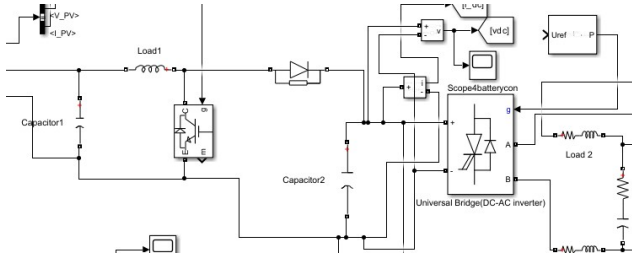


Figure 30: Load and capacitor connected to the MPPT PV array-boost converter and the input of the inverter: Load 1, capacitor 1 and Capacitor 2. In the Simulink model.

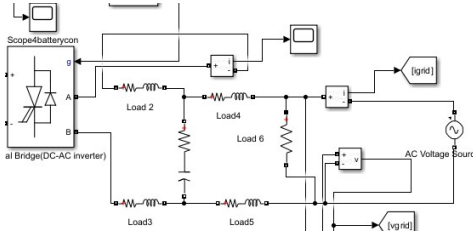


Figure 31: Grid-connected loads 2,3,4,5,6. In the Simulink Model

13.5 Load

The load is presented by RLC branch. A few capacitors are used in the circuit to ease the DC-AC conversion. The parameters resistance R , inductance L and capacitance of C of each load and capacitor is in [16]. From left to right, capacitor 1 and load 1 are connected to the output of the PV array and the input of the booster. The output of the boost converter, before entering the DC-AC inverter universal bridge, is connected to yet another capacitor 2. See figure 30. The initial voltage values of capacitor 1 and 2 are 200V and 435 V, respectively.

From the output of the DC-AC universal bridge converter, there are several grid-connected parallel loads: Loads 2, 3, 4, 5, 6. See figure 31. The R, L , and C values of each load and capacitor are in table 1.

The values of the loads and capacitors are listed in Table 1.

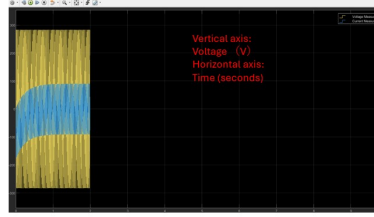
Table 1: Load/Capacitor values from [16]

Component	Resistance(Ω)	Inductance(L)	Capacitance(F)
Load 1		3e-3	
Capacitor1			1000e-6
Capacitor2			3000e-6
Load 2	8.2e-3	2.18e-3	
Load 3	8.2e-3	2.18e-3	
Load 3	8.2e-3	2.18e-3	
Load 4	8.2e-3	2.18e-3	
Load 5	8.2e-3	2.18e-3	
Load 6	8.2e-3	2.18e-3	

13.6 Grid-tie inverter

The grid-tie inverter is represented in Simulink as a Universal bridge.

• If isolated from the battery



If connected to the battery

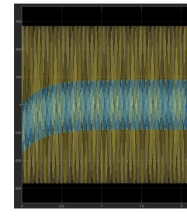


Figure 32: V and I of the grid connected part of the circuit

13.7 Grid

The grid is represented by an AC source voltage. Its peak amplitude voltage is $200\sqrt{2}$ V, and the frequency is 60 Hz[17].

13.8 Some Issues of the Simulation Model

- **Battery and Power issues:** In our initial result, the graph of V and I of the grid measured against time (in seconds) remains constant, regardless of whether the microgrid is connected to the battery or not. The yellow lines represent voltage, and the blue lines represent current. The maximum voltage value is almost 300V, and the maximum current value is almost 100 A. As can be seen in the figure 32, the graph of PV microgrid without battery and the PV microgrid with the battery is identical. Both the V-I graph of the battery-connected microgrid and not connected to the battery-microgrid is simulated in 2 seconds. The constant V-I graph for the battery and grid-connected microgrid is unusual as usually $P_{grid} = P_{PV} + P_{grid}$ ($I_{grid} = I_{PV} + I_{battery}$). This may indicate the unusual configuring of the grid-connected load and grid, and the battery seems to work incorrectly.
- The power analysis of this model is also incomplete as we have only considered the voltage and current graph, and have not yet measured all power graphs of the MPPT-controlled PV array, battery, and grid. The result of the MPPT power (not shown here) is improperly measured, $P_{PV}=0$, as Simulink does not measure I_{PV} due to capacitance values. This may be due to improper placement of current measurement block used to measure current output of the MPPT-controlled PV array. Other than the power issues discussed in the previous section of the model, some issues include:
- Very slow simulation done on the computer, using either my laptop or the computer in the lab, even though we only simulate parameters for less than 10 seconds. This may indicate slow sampling time. Slow sampling time is an issue if we decide to incorporate more complex microgrid systems with more components. Therefore, the sampling rate should be changed.

13.9 Resolution

The resolution will be to rebuild model from scratch, change sampling rate, and put proper configuration of the current measurement block and voltage measurement.

This is my initial first time modelling a PV-battery microgrid. There are still many errors and the constructed model

is very poor. However, this is a valuable experience and I can gain insight into what to avoid and potential new methods to improve the model.

14 Conclusion and Future Research Plan

First, basic categories of microgrid were discussed. Then, a literature survey was done to learn some types of microgrid control and renewable energy- integrated microgrid challenges. Renewable energy system (RES) integrated into the power system is an important solution to help mitigate environmental and sustainability concerns. However, RES has challenges due to its intermittency, which can lead to unstable operation of the microgrid.

Before we design the microgrid, we read literature about solar-battery microgrid and hydrogen-based microgrid. Then, I attempted to model in Simulink a simple microgrid consisting of a PV array, battery, load, and grid represented as an AC source. The model itself was not a success, but it gave a valuable initial learning experience. If the model is successful, hydrogen storage system will be added to the microgrid later.

In the future, we are trying to find a specific research question regarding the renewable energy microgrid. First, we will consider which scale of the microgrid to focus on: lower scale (individual microgrid) which focuses on voltage/current, P/Q control etc, or higher system-scale which focuses on only the power exchange between 1 or more microgrids. In PBL II, we will begin to consider control methods to tackle specific research problems. Then, simulation will be done on Simulink MATLAB to test the control methods.

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